

PRESS STUD (SNAP FASTENER) IN PCBA ASSEMBLY

Complete Guide to Types, Assembly Methods & Applications in Electronics

Topics Covered:

- What is a Press Stud / Snap Fastener
- Types: Metal, Plastic, Broaching, Snap-Top
- PCB Assembly Methods: SMT, Press-Fit, Broaching
- Applications in Electronics & Medical Devices
 - Design Considerations & DFA Rules
 - Common Challenges & Solutions

1. WHAT IS A PRESS STUD (SNAP FASTENER)?

A **press stud** (also called **snap fastener**, **snap button**, or **popper**) is a pair of interlocking discs typically made of metal or plastic, used to fasten two surfaces together by pressing them until an audible 'snap' is heard. In PCBA assembly, press studs serve as **quick-release mechanical fasteners** for:

- **PCB-to-Enclosure Attachment:** Quick mounting and removal of PCBs from housings without screws
- **Board-to-Board Stacking:** Temporary or modular connections between daughterboards and motherboards
- **Medical Electrode Connections:** ECG/EEG snap buttons for patient monitoring devices (3.5mm, 4.0mm standard)
- **Test Fixture Access:** Rapid opening/closing of test jigs and programming fixtures
- **Battery Compartment Covers:** Consumer electronics where frequent battery replacement is needed
- **Shielding & Grounding:** Metal press studs providing both mechanical retention and electrical continuity

In electronics manufacturing, press studs are classified into two main categories: **metal snap fasteners** (brass, stainless steel, phosphor bronze) for high-strength and conductive applications, and **plastic snap fittings** (nylon, POM, ABS) for insulated, lightweight, and cost-sensitive applications. The choice depends on mechanical load, electrical requirements, environmental conditions, and assembly method.

2. TYPES OF PRESS STUDS IN PCBA

Press studs used in PCBA assembly come in various designs optimized for specific mounting methods and functional requirements.

Type	Description	Material	Assembly Method
Broaching Snap-Top Standoffs	Knurled-shank fasteners with spring-action top that	Brass, Steel, Stainless Steel, Aluminum	Broaching (Press-fit into PCB hole)
Snap-Fit Plastic Spacers	Nylon or POM plastic posts with barbed or harpoon	Acrylic, PPS, PMMA, PC, PPS, PPS	Press-fit (Insert and press)
ECG / Medical Snap Buttons	Small metal snap fasteners (3.5mm, 4.0mm) for electrode	Stainless Steel, Brass, Titanium	Press-fit (Insert into 5.0mm hole)
Press-Fit Power Elements	Heavy-duty metal studs with compliant pins pressed	Titanium, Brass, Copper	Press-fit (Insert into M3/M4/M5/M6/M8/M10 threads)
SMT Snap Fasteners	Surface-mount metal or plastic fasteners with flat base	Steel (plated), Aluminum, Plastic	Reflow (Solder onto PCB)
Self-Clinching Snap Studs	Metal studs that clinch into PCB or chassis material	Stainless Steel, Brass, Aluminum	Clinching (Press into pre-drilled hole)
Quick-Release Broaching Standoffs	Spring-action standoffs that hold panels securely when	Brass, Steel with plated brass	Broaching (Press into hole)
Flare-Mounted Snap Standoffs	Broaching standoffs with flared shank for enhanced	Brass (C360), Spring Steel	Broaching (Press into hole)

Image Reference — Standard Press Stud / Snap Fastener Types:



Various metal press stud / snap fastener sizes and finishes for industrial and textile applications

Image Reference — Industrial Snap Fastener (DOT Type):



DOT-type durable snap fastener press stud with cap, socket, stud, and post components

3. ASSEMBLY METHODS FOR PRESS STUDS IN PCBA

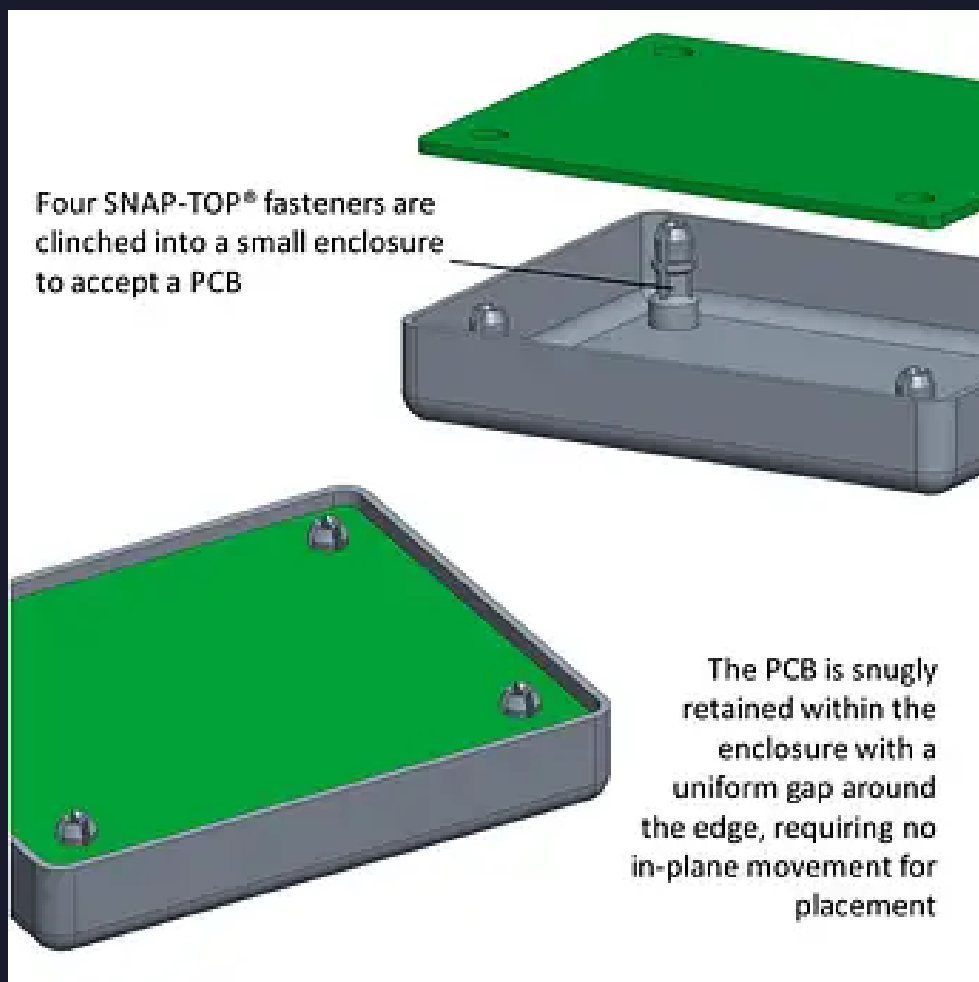
Press studs can be assembled onto PCBs using multiple technologies depending on the stud type, material, and production requirements.

3.1 Broaching (Press-In) Method

Broaching is the most common method for metal press studs and standoffs in PCBA. The fastener has a **knurled or serrated shank** with axial grooves that cut into the PCB material when pressed into a drilled hole, creating a permanent interference fit.

- **Installation:** Place fastener in anvil hole → align PCB mounting hole over shank → apply parallel squeezing force until shoulder contacts board.
- **Tools:** Flat punch and recessed anvil. For flare-mounted types, a punch flaring tool flares the extended shank outward for enhanced retention.
- **Hole Requirements:** Non-plated holes only for broaching (plated holes may crack). Hole diameter tolerance: $\pm 0.05\text{mm}$. Minimum panel thickness: 1.27mm (FR4).
- **Materials:** Suitable for non-ductile materials: FR4 epoxy, polycarbonate, acrylic, glass epoxy, resin laminates. Max panel hardness: 60 HRB.
- **Advantages:** Permanent installation, no solder/heat, no loose hardware, resistant to rotation and push-out.

Image Reference — Snap-Top Standoffs (Broaching Type):



PEM SNAP-TOP standoffs — broached into enclosure with spring-action snap retention for PCB



PEM SNAP-TOP standoff close-up showing broached shank and snap-action retention design

3.2 Press-Fit Technology

Press-fit studs (also called **Power Elements**) use compliant pins that deform when pressed into plated through-holes, creating a gas-tight electrical and mechanical connection without solder.

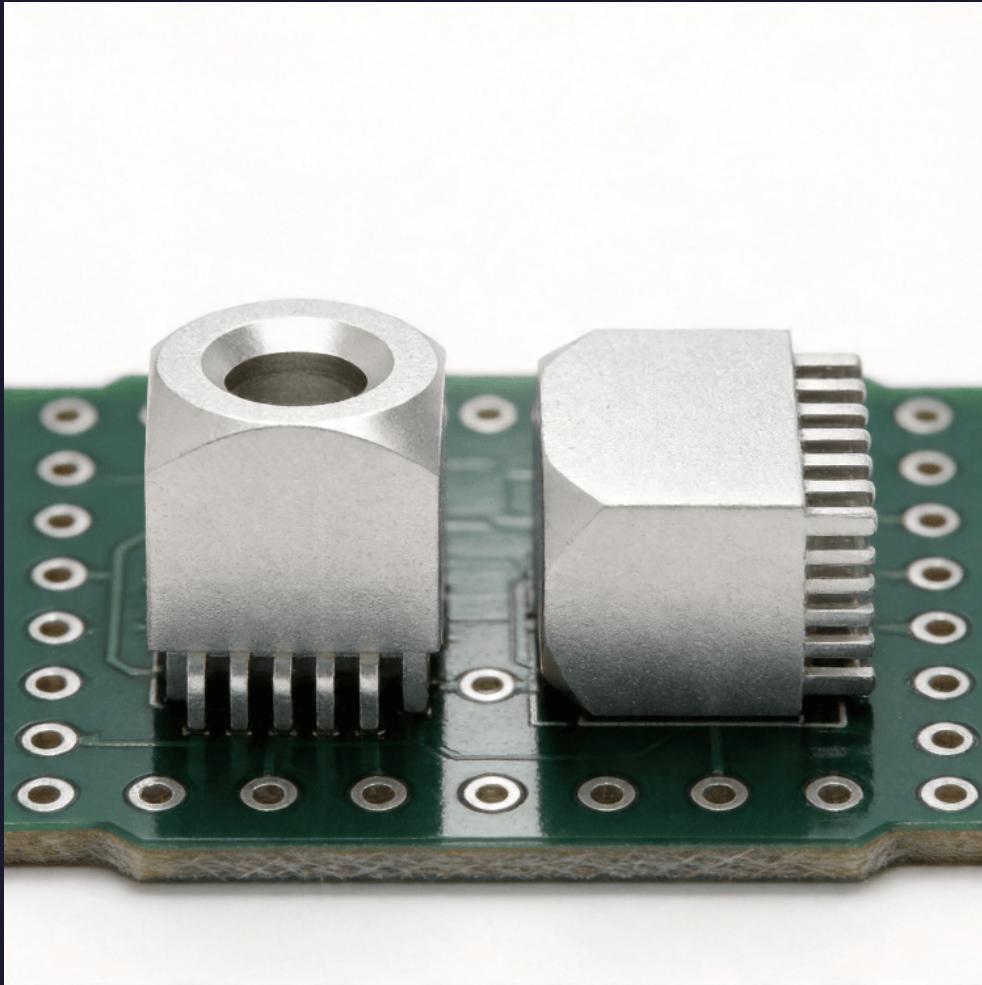
- **Pin Types:** Compliant pins (flexible, lower insertion force) or solid pins (higher retention, used for high-current).
- **Insertion Force:** 60–250N per pin depending on size (M3 to M10). Total fixture force must be calculated for multi-pin arrays.
- **Hole Specs:** Plated holes with 25µm minimum copper, 15µm maximum tin. Hole diameter: 1.0mm $\pm 0.09/-0.06$ for standard pins.
- **Reusability:** Press-fit connectors can typically be removed and replaced up to 3 times without damaging the PCB.
- **Applications:** High-current power distribution, automotive, solar equipment, industrial controls.

Image Reference — Press-Fit Power Elements:

Sandum™



M3/M4/M6 press-fit power elements with threaded tops and compliant press-fit pins for PCB mounting



Press-fit PCB connectors with M8/M10 threads mounted on PCB showing compliant pin array

3.3 Surface Mount Technology (SMT)

SMT press studs and fasteners are supplied on tape-and-reel for automated pick-and-place assembly. They have a flat base with solderable pads that reflow along with other SMD components.

- **Pick-and-Place:** Pneumatic fingers require a flat suction surface. Some fasteners need a Kapton/polyimide patch for vacuum pickup, removed after reflow.
- **Reflow Profile:** Standard lead-free reflow (245–260°C peak). Tin-plated steel fasteners solder reliably.
- **Panel Fasteners:** Consist of a soldered retainer base + snap-on cap/screw assembly. The retainer goes through reflow; cap attaches afterward.
- **Right-Angle Fasteners:** SMT fasteners with two pilot pins for stability. Rectangular hole reduces thermal mass for faster heating.
- **Advantages:** Single-pass assembly, no secondary operations, compatible with standard SMT lines.

3.4 Snap-Fit Plastic Method

Plastic snap-fit studs and spacers use barbed or harpoon-shaped ends that compress during insertion and expand to lock behind the PCB surface.

- **Installation:** Hand-pressed or light pneumatic press. No heat or solder required.
- **Materials:** Nylon 6/6 (most common), POM (Delrin), ABS, polycarbonate. Nylon offers good strength and insulation.

- **Hole Requirements:** Non-plated holes, diameter matched to barb size (typically 3.0–4.0mm). Panel thickness: 1.0–3.0mm.
- **Retention:** Barbs grip the back side of the PCB. Pull-out force: 20–80N depending on design and material.
- **Applications:** Consumer electronics, low-cost assemblies, insulated standoffs, temporary fixtures.

Image Reference — Snap-Fit Plastic PCB Supports:



Nylon PCB support snap-in latch with harpoon end spacer for plastic snap-fit assembly

4. MEDICAL SNAP BUTTONS (ECG / EEG / EKG)

Medical-grade snap buttons are a specialized category of press studs used extensively in patient monitoring devices, TENS/EMS units, and diagnostic equipment.

4.1 Specifications

Parameter	Standard Value	Notes
Diameter	3.5mm or 4.0mm	3.5mm most common; 4.0mm for heavy-duty
Material	Nickel-plated Brass or Stainless Steel	Nickel plating prevents corrosion from skin contact
Mounting	PCB Mount Female Socket	2-claw or 4-claw retention on PCB
Mating Part	Male Snap Stud on Lead Wire	Stainless steel 3.9mm typical
Current Rating	1–5A	Low current — signal path only
Contact Resistance	<10mΩ	Critical for signal integrity in medical apps
Cycles	5,000–10,000 mate/demate	Higher than consumer-grade snaps
Standards	ISO 13485, IEC 60601	Medical device quality requirements

4.2 PCB Mounting Methods

- **2-Claw PCB Mount:** Two metal claws grip the PCB edge or through-hole. Simple, low-cost, but lower retention force.
- **4-Claw PCB Mount:** Four claws provide 360° grip. Better retention and stability. Used in portable devices.
- **Solder Tail Mount:** Extended solder tails pass through PCB holes and are wave or hand soldered. Most secure method.
- **SMT Mount:** Flat base with solderable pads for reflow. Enables automated assembly but requires larger PCB footprint.

4.3 Applications

- **ECG/EKG Machines:** 10-lead patient cables with snap connectors
- **TENS/EMS Units:** Electrode lead connections for pain therapy devices
- **EEG Systems:** Electrode caps with snap-button sensor attachments
- **Defibrillators:** Quick-connect electrode pads for emergency use
- **Wearable Health Monitors:** Fitness bands and smart patches with snap-in sensors

5. DESIGN FOR ASSEMBLY (DFA) CONSIDERATIONS

5.1 Hole Size & Tolerance

Proper hole sizing is critical for reliable press stud installation:

Stud Type	Hole Diameter	Panel Thickness	Tolerance
Broaching Snap-Top (Small)	2.54mm	1.27–1.6mm	±0.05mm
Broaching Snap-Top (Standard)	3.58–4.0mm	1.27–3.2mm	±0.08mm
Broaching Snap-Top (Large)	5.41–6.35mm	1.6–4.0mm	±0.13mm
Snap-Fit Plastic (Small)	3.0mm	1.0–2.0mm	+0.1/-0.0mm
Snap-Fit Plastic (Standard)	4.0mm	1.5–3.0mm	+0.1/-0.0mm
Press-Fit Power (M3)	1.0mm (plated)	1.6–3.2mm	+0.09/-0.06mm
Press-Fit Power (M4)	1.2–1.4mm (plated)	1.6–4.0mm	+0.1/-0.08mm
Medical Snap (3.5mm)	3.5mm socket mount	1.0–2.0mm	±0.05mm

5.2 Edge Distance & Clearance

- **Minimum Edge Distance:** 2.54mm (0.1") from hole center to PCB edge for broaching fasteners. Closer distances risk panel cracking.
- **Component Clearance:** 3–5mm keep-out zone around press studs to allow for installation tooling access and to prevent interference with tall components.
- **Board Thickness:** Broaching requires minimum 1.27mm FR4. Thinner boards (0.8mm) require SMT or snap-fit plastic types instead.
- **Hole-to-Hole Spacing:** Minimum 1.5× hole diameter center-to-center to prevent material tearing between adjacent studs.

5.3 Material Compatibility

Material selection affects both the fastener and the PCB:

- **FR4 Epoxy:** Standard for broaching and snap-fit. Hardness must be <60 HRB for broaching to cut properly without cracking.
- **Metal Chassis (Steel/Aluminum):** Use self-clinching or grounding standoffs. Broaching is NOT suitable for ductile metals.
- **Plastic Enclosures:** Snap-fit plastic studs or threaded inserts. Avoid broaching in soft plastics (ABS, PP) — use heat-stake inserts instead.
- **Plating Considerations:** Tin-plated fasteners solder well. Nickel-plated fasteners resist corrosion but may require higher soldering temperatures.

5.4 Electrical & Grounding

When press studs serve as electrical connections:

- **Grounding Standoffs:** Use SOAG/SOSG series with gripping teeth that bite into chassis metal for firm electrical contact.
- **Contact Resistance:** Press-fit connections: <0.5mΩ. Soldered connections: <1mΩ. Measure with 4-wire Kelvin method.

- **Current Capacity:** Single M3 press-fit pin: ~15A. For higher current, use multiple parallel pins or M6/M8 power elements.
- **EMI Shielding:** Metal snap studs can connect PCB ground planes to enclosure shields. Ensure low-impedance path.

6. COMMON CHALLENGES & SOLUTIONS

Challenge 1: PCB Cracking During Broaching

Excessive squeezing force or too-small edge distances cause FR4 laminate to crack around the fastener hole.

Solutions:

- Maintain minimum 2.54mm edge distance (check manufacturer spec)
- Use proper anvil and punch tooling — do not over-squeeze
- For thin boards (<1.27mm), switch to SMT or snap-fit plastic types
- Pre-test on sample boards to verify hole size and force requirements

Challenge 2: Loose Snap-Fit Plastic Studs

Plastic snap studs can loosen over time due to thermal cycling, vibration, or material creep (especially nylon at elevated temperatures).

Solutions:

- Use POM (Delrin) instead of nylon for better dimensional stability
- Specify barbed designs with multiple retention ridges
- Add epoxy staking around the stud base for critical applications
- Design hole diameter at the lower end of tolerance for tighter fit

Challenge 3: Press-Fit Pin Damage to Plated Holes

High insertion force can scrape plating from hole walls, especially with solid (non-compliant) pins.

Solutions:

- Use compliant pins (Eye of the Needle, Action Pin, C-Press designs)
- Verify hole plating thickness $\geq 25\mu\text{m}$ copper
- Limit re-insertions to 3 cycles maximum
- Use insertion tools with ejectors for safe removal

Challenge 4: SMT Fastener Co-Planarity Issues

SMT press studs may not sit flat due to solder paste thickness variations or warped boards, leading to poor solder joints.

Solutions:

- Use planar PCB finishes (ENIG, Immersion Silver) — avoid HASL
- Control solder paste thickness to $150\mu\text{m} \pm 25\mu\text{m}$
- Design fastener pads 0.5mm larger than the component footprint
- For threaded fasteners, use a Kapton patch for vacuum pickup stability

Challenge 5: Galvanic Corrosion in Mixed-Metal Assemblies

When brass studs contact aluminum chassis or steel screws in humid environments, galvanic corrosion accelerates.

Solutions:

- Use compatible plating (tin on both surfaces, or nickel barrier)
- Apply conformal coating over fastener joints
- Use stainless steel studs for marine/outdoor applications
- Design drainage paths to prevent moisture accumulation

Challenge 6: Medical Snap Button Signal Integrity

Poor contact between medical snap socket and stud can introduce noise in sensitive bio-signal measurements.

Solutions:

- Specify gold or nickel plating with $<10\text{m}\Omega$ contact resistance
- Use 4-claw PCB mounts for better mechanical stability
- Design guard rings around snap connections to reduce leakage
- Implement shielded cable assemblies with snap connectors

7. REFERENCE IMAGES

Press-Fit PCB Connectors (M3/M4/M5 Power Elements):



M3/M4/M5 PCB connector press-fit power element with threaded top and compliant pins

Variety of Press-Fit PCB Terminals:

Sandum™



Collection of press-fit PCB terminals, nuts, spacers, and power elements in various sizes

Snap-Fit Press-In Metal Bolts:



Brass snap-fit press-in metal bolts with knurled retention and split-shank design for PCB mounting

8. QUICK REFERENCE — PRESS STUD SELECTION GUIDE

Use this guide to select the appropriate press stud type for your PCBA application:

Application	Recommended Type	Key Specs
Quick PCB mounting/removal in enclosures	Broaching Snap-Top Standoff	Brass, tin-plated, spring-action, tool-free
High-vibration automotive/industrial	Press-Fit Power Element	Compliant pins, M3–M10 thread, 60–250N insertion
Medical electrode connections (ECG/EEG)	PCB Mount Snap Button	3.5mm/4.0mm, nickel-plated brass, 4-claw mount
Low-cost consumer electronics	Snap-Fit Plastic Spacer	Nylon 6/6, barbed end, hand-press install
Automated SMT assembly line	SMT Snap Fastener / Panel Screw	Tape-and-reel, tin-plated steel, reflow compatible
PCB-to-chassis grounding	Grounding Standoff (SOAG)	Gripping teeth, steel, clinch into metal chassis
High-current power distribution (50A+)	Press-Fit Power Element (M6+)	Multiple compliant pins, tin-plated brass
Temporary test fixture / programming	Snap-Top Standoff or Quick-Release Broach	Spring-action, repeated snap-in/out cycles
Board-to-board stacking (modular)	Broaching Standoff (Threaded)	M3/M4 internal thread, permanent press-in mount

Key Takeaways:

1. Press studs in PCBA serve as quick-release mechanical fasteners for PCB mounting, medical connections, and modular assemblies.
2. Broaching snap-top standoffs are ideal for tool-free PCB removal in enclosures — use brass/tin-plated types for conductivity.
3. Press-fit power elements eliminate soldering for high-current paths — specify compliant pins to protect PCB plating.
4. Medical snap buttons (3.5mm/4.0mm) require nickel-plated brass and 4-claw PCB mounts for reliable bio-signal integrity.
5. Always verify hole size, edge distance, and panel thickness against manufacturer specifications before specifying press studs.
6. For automated assembly, choose SMT-compatible fasteners on tape-and-reel with Kapton patches for vacuum pickup.